

Databases organize information into structured tables with rows and columns that relate data. Queries retrieve and manipulate data based on specified conditions and parameters. Each table has a unique name within a schema and constraints enforce data integrity by rejecting inserts and updates that violate defined rules. Indexes are auxiliary data structures that allow efficient retrieval of data. Transactions group multiple statements into a single atomic unit of work, ensuring consistency and preventing conflicts.

The first section introduced tables as the primary unit of storage. Relational databases use the same SQL syntax across the database. The second section covered indexes for performance. Foreign keys reference rows in other tables to enforce referential integrity. The third section will explain joins across multiple tables. Materialized views precompute expensive aggregations to accelerate queries.